

ACKNOWLEDGEMENT

UBC is located on the traditional, ancestral, and unceded territory of the Coast Salish Peoples, including the territories of the xʷməθkʷəy̓əm (Musqueam), Skwxwú7mesh (Squamish), Stó:lō and Səlilwətaʔ/Selilwitulh (Tsleil-Waututh) First Nations. We are mindful of our responsibility to reconciliation given our position on this stolen land, and are committed to upholding this responsibility with humility and openness to feedback.



Figure 2. *Raven Brings The Light* (by Haida artist and former UBC student, Bernard Kerrigan)

We acknowledge our position as one part of a colonialist educational institution that has been used as a means of assimilating Indigenous peoples, leading to trauma that echoes across generations. That colonialism persists in structural, institutional, and personal colonialism and racism, impacting the rights and resources of Indigenous persons, their languages, their art, and their cultures.



Figure 1. *sʔi:ʔqəy̓ qeqən: Musqueam Post* (by Musqueam artist, Brent Sparrow Jr.)

COURSE INFORMATION

Course Title	Course Code Number	Credit Value
Neuroaesthetics: <i>Your Brain on Art</i>	ASTU_V 400E	3

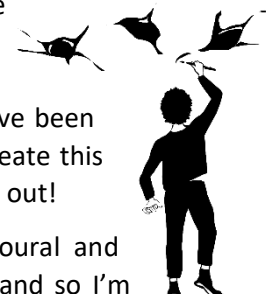
INSTRUCTOR CONTACTS

Course Instructor(s)	Contact Details	Office Location	Office Hours
Leo J. Chung	ljchung@student.ubc.ca	DMCBH 4502A	Reach out to schedule office hours
Betty Bao	bettybao507@gmail.com	TBD	Reach out to schedule office hours

Course Mentor	Contact Details	Office Location	Office Hours
Steven J. Barnes	sjb@psych.ubc.ca	CIRS 4344B	N/A

COURSE COORDINATOR BIOGRAPHICAL STATEMENTS

My name is Leo, and I am a junior studying Cellular Neuroscience at UBC who was born and raised in Vancouver. After transferring from the University of Washington following my freshman year, I began studying the impact of inflammation and immune responses on microglial and gene expression in the brain with Dr. Annie Ciernia at the Djavad Mowafaghian Centre for Brain Health. Currently, my work in the lab lies in uncovering the role of BAF53b – a chromatin-remodelling complex – on neurodevelopment. Outside of academia, I have been interested in the brain's role in hedonics, which has inspired me to co-ideate and create this seminar; *Neuroaesthetics: Your Brain on Art*, with Betty and Steven. Feel free to reach out!



Hello! my name is Betty and I'm a 4th year Neuroscience student in the Behavioural and Cognitive Stream at UBC. I've been involved in various interdisciplinary initiatives, and so I'm really interested in the connection of neuroscience across disciplines and communities, which is

sparked from my interests in painting, music, and comics. Currently, I'm involved in the Van Lab under Dr. VanHandel of the Music Department, studying the influence of musical experience on the perception of musical complexity. Recently, I've also gained an appreciation for zine making and Nikolai Kapustin's concert etudes. If you have any questions about anything, or just want to talk, email me!

COURSE DETAILS

Course Learning Outcomes: We hope that by the end of this course, you will be able to:

LO1. Evaluate key concepts in neuroaesthetics by reflecting on how perceptual, emotional, and cognitive processes shape aesthetic experiences across various art forms and be able to identify how this applies in their personal life.

LO2. Analyze personal and cultural aesthetic experiences using neuroaesthetic principles and explain and communicate to an audience on how those principles shape individual perception and taste.

LO3. Evaluate real-world applications of neuroaesthetics (e.g. music therapy, museum design, art-based interventions) and their impact on cognition and well-being.

LO4. Create and evaluate an original art piece informed by a specific aesthetic or perceptual principle, accompanied by a thoughtful artist statement that explains the conceptual and experiential basis of the work.

LO5. Gain an inclusive and culturally sensitive understanding of neuroaesthetics through analysis and discussion of readings and perspectives from guest speakers.



Course Structure: This course consists of seminar-style classes twice a week, which will consist of in-class activities and discussions, and occasionally guest lectures. The location of the classes will be in [BIOL 1001](#), which is on the first floor of the Biological Sciences Building. None of the classes will be live-streamed or recorded. Lecture slides will be available on Canvas under Modules before each class to facilitate your preparation. There is also freedom and flexibility for students to shape the course and their learning; together as a class, you can vote on topics to focus on and can provide input at any time through anonymous feedback forms to help develop and improve the course throughout the semester.

Class Structure: For each class, there is a large emphasis on participation and engagement, and you are expected to actively engage with the content and discussions, as your participation will shape the course and each other's learning. At the beginning of the semester, you will discuss with other students and co-create a community agreement, which serves to establish a safe and welcoming learning environment. Class content will be largely based on readings, which should be completed before each class. Most classes will consist of a mixture of reflections, discussions, and group activities.

LEARNING ACTIVITIES

You are expected to attend each of the weekly class meetings and participate actively. Classes will include class-wide discussions, small-group activities, presentations, and some lecturing regarding seminar content. Participation in discussions and activities are required for maximum achievement of the participation grade in the seminar. Assigned readings and pre-class work should be completed in advance of the seminar time for which it is assigned for. Assignments throughout the course are expected to be completed on time and are mandatory. Class time will be given for certain smaller assignments, while larger-scale assignments will require you to complete that work outside class time.

SCHEDULE OF TOPICS (MAY BE SUBJECT TO CHANGE)

Dates	Topic(s)	Readings* (Assigned pages to be complete before class)	Due Dates
Jan 6 th	<i>Introduction to the course and class expectations</i> <i>Introducing yourselves</i> <i>Collaboratively establishing a learning agreement</i> <i>Introduction to the study of neuroaesthetics.</i>		Opening reflection (1%)
Jan 8 th	<i>History of neuroaesthetics</i> <i>Discussion of future course topics and student presentations</i>	Skov, M. (2022a). Neuroaesthetics as a scientific discipline. The Routledge International Handbook of Neuroaesthetics (p. 1-7)	
Jan 13 th	<i>The neurobiology of liking</i> <i>How to critically reflect and communicate</i>	Skov, M. (2022b). The neurobiology of liking. The Routledge International Handbook of Neuroaesthetics (p. 63-68)	Aesthetic experience reflection (2%)
Jan 15 th	<i>The neurobiology of disliking</i> <i>Effective peer assessment and marking</i>	Klebl, C. et al. Disliking: from adaptive disgust to ugliness. The Routledge International Handbook of Neuroaesthetics (p. 78-81)	
Jan 20 th	<i>Individual differences and universality</i>	Nadal, M., & Chatterjee, A. (2018). Neuroaesthetics and art's diversity and Universality.	
VISUAL ART AND THE BRAIN			
Jan 22 nd	<i>Introduction to visual art, visual perception and related brain areas.</i>	Ramachandran, V., & Hirstein, W. (1999). The Science of Art: A Neurological Theory of Aesthetic Experience.	

	<i>The eight laws of artistic experience (Ramachandran)</i> <i>Overview of Assignment 1: Creative Piece and Artist Statement + Critique</i>		
Jan 27 th	<i>Theories and models in neuroaesthetics</i> <i>Problems and controversies with neuroaesthetics approaches</i>	Neidich, W. (2022). An Activist Neuroaesthetics Reader. (p. 6-16, 29-30)	
Jan 29 th	<i>GUEST SPEAKER: Brain injury and dysfunction and its impact on artistic production</i>	(OPTIONAL READING) Sherwood, K. (2012). How a Cerebral Hemorrhage Altered My Art.	
Feb 3 rd	<i>Psychoactive substances and alcohol on creativity</i>	Fokt. S. (2024). High Art, High Artists.	
Feb 5 th	<i>The influence of physical context and knowledge of artworks on aesthetic experience (Experiencing art in museums/social settings)</i> <i>Overview of the Neuroaesthetics Art Gallery</i>	Illes, A., & Tinio, P. P. L., (2022) Experiencing art in museum. The Routledge International Handbook of Neuroaesthetics. (p. 423-433)	
Feb 10 th	<i>How to critique effectively</i>		Unit reflection (2%)
Feb 12 th	<i>Neuroaesthetics Art Gallery: Presenting your Creative Piece and Artist Statement</i>		Creative Piece and Artist Statement + Critique (20%)
Feb 17 th	READING BREAK		
Feb 19 th	READING BREAK		

MUSIC AND THE BRAIN			
Feb 24 th	<i>Introduction to music and the brain</i>	Belfi, A. M. & Loui, P. (2022). The music system. The Routledge International Handbook of Neuroaesthetics (p. 315)	Aesthetic experience reflection (2%)
Feb 26 th	<i>Music and emotion</i>	Taruffi, L. & Koelsch, S. (2022). Music-evoked emotions : their contribution to aesthetic experiences, health and well-being. The Routledge International Handbook of Neuroaesthetics	
Mar 3 rd	<i>GUEST LECTURE: Cultural differences in music tastes</i>	Berger, K. (2021). What Makes Music Universal. Nautilus. https://nautilus.us/what-makes-music-universal-238187/	
Mar 5 th	<i>Brain of the performing artist</i>	Ullen, F. (2022). Expertise and the brain of the performing artist. The Routledge International Handbook of Neuroaesthetics	Unit reflection (2%)
Mar 10 th	Student Presentations		
ART AND BRAIN WELLNESS			
Mar 12 th	<i>Neuroaesthetic Topics in Wellness: Applications of neuroaesthetics in health, wellness and society (music/art therapy)</i> <i>Overview of Infographic Assignment</i>	Howlin, C. (2022). The health benefits of art experience. The Routledge International Handbook of Neuroaesthetics	
Mar 17 th	<i>Neuroaesthetics & Wellness: (cont'd) Applications of neuroaesthetics in health, wellness and</i>	de Paiva, A., & Jedon, R. (2019). Short- and long-term effects of architecture on the brain: Toward theoretical formalization.	Unit reflection (2%)

	<i>society (dementia/PTSD accessible architecture and urban design)</i>		
Mar 19 th	<i>Student Presentations</i>		Infographic Assignment (25%)
OTHER TOPICS IN NEUROAESTHETICS			
Mar 24 th	<i>Odour/gustation aesthetics: hedonic perception of olfactory and taste stimuli</i>	Dikecligil G. N. & Gottfried J. A. (2022). Odour aesthetics: hedonic perception of olfactory stimuli. The Routledge International Handbook of Neuroaesthetics	Aesthetic experience reflection (2%)
Mar 26 th	<i>Literature, Poetry and Narrative</i>	Jacobs, A. M. (2022). Literature and poetry. The Routledge International Handbook of Neuroaesthetics Hartung, F. (2022). Narrative. The Routledge International Handbook of Neuroaesthetics, 410–422.	
Mar 31 st	<i>Student Presentations</i>		
Apr 2 nd	<i>Student Voted Topics</i>	The Routledge International Handbook of Neuroaesthetics	
Apr 7 th	<i>Student Presentations</i>		
Apr 9 th	<i>Ways to continue exploring neuroaesthetics + Reflection of learning</i>	Ahtone, H. (2019). Considering Indigenous Aesthetics: a non-Western paradigm.	End of term reflection (1%)
END OF TERM			

**Note: You will be given the chance to vote for topics and associated readings. For classes after reading break, there is a chance that the syllabus will be subject to change. This changes will be mentioned far ahead in class with enough time for preparation.*

LEARNING MATERIALS

There will be a list of readings provided at the beginning of the course, which will be the basis for the course content and our discussions. All the readings will be available to be accessed online. A large portion of our course is based on the textbook: The Routledge International Handbook of Neuroaesthetics (Martin Skov, Marcos Nadal). This textbook can be accessed for free online using your UBC account. It is also possible to purchase a physical version of the textbook for \$44.42 USD if desired.

You will have access to the reading list, the slides, and submit assignments through Canvas. There will be pre-made modules on Canvas to supplement and build fundamental neuroscience knowledge, if necessary.

ASSESSMENTS OF LEARNING

Reflections (15%)

You will be writing a series of reflections throughout the course. Reflections will be peer-marked, using specific rubrics, and any citation style is acceptable as long as it is consistent throughout the assessment. Reflection assessments will include:

Opening Reflection (1%) (200-250 words)

At the beginning of the course, you will be asked to reflect on your prior experiences with art, neuroaesthetics, or hedonics; and/or your expectations for this class.

End-of-course Reflection (2%) (200-250 words)

During the final class of the term, you will compose a second reflection piece that focuses on your time in the course and your key take-aways. You can choose to reference your reflection from the beginning of the course if it makes sense for the content of your letter.

Aesthetic Experience Reflections (3 total, 6%, each 2%)

You will answer a question related to the concept of the week in relation to an artwork (painting, song, short film) displayed at the beginning of first class of each weekly theme. (200-250 words).

Can miss one out of the three without consequence. Weight will shift to the other two reflections, resulting in each reflection being worth 3%.

Unit Reflections (3 total, 6%, each 2%)

After each unit, you will complete a short reflection on one idea or concept they learned and how it applies to their own life or worldview. (100-150 words)

Can miss one out of the three without consequence. Weight will shift to the other two reflections, resulting in each reflection being worth 3%.

Together, these reflections support the seminar's goals of fostering open-minded exploration by centering individual perspectives alongside course content.

Creative Art Piece and Artist Statement + Critique (20%)

You will produce a creative work (different visual art forms will be accepted) inspired by 2-3 perceptual or aesthetic principles studied in class (e.g. contrast, symmetry, art as a metaphor, etc.). You can choose to either abide by or break these principles to achieve a certain aesthetic effect. Alongside the

art piece, students will submit an artist statement (approx. 500–750 words). You will not be marked on the quality of the art piece, rather you will be evaluated based on the artist's statement, which will explain the perceptual principles, the rationale behind your artistic choices, the aesthetic effect you tried to achieve, your reflection on the validity of the principles and how the work relates to course concepts. Afterwards, you will provide peer feedback for one student's artist statement in the form of a written critique (200 words). The peer feedback will be graded for participation.

Infographic Creation (25%)

After learning about some of the applications of neuroaesthetics in different contents, students will research an applied area of neuroaesthetics (e.g., music therapy, art-based interventions, museum design, etc.) and create an infographic that introduces and explains the topic to a general audience. The final infographic will be posted publicly on Canvas for peer assessment.

Student-Led Presentation and Discussion Project (30%)

Students will deliver a 15-20 minute presentation in pairs, focusing on a topic within the neuroaesthetics.

After the presentation, the students will lead a discussion on the broader ideas of their selected topic. The students should prepare 3-4 discussion questions to guide the conversation. At the end of the presentation, the students of the class will provide feedback for the presentation using specific rubrics, providing both constructive and encouraging comments. The entire presentation and discussion should take up to 35 minutes, and students are free to decide how much time they want to dedicate to the presentation vs discussion portion.

These presentations will occur in groups of 2-3. Two groups in each presentation slot, throughout the entire seminar, with groups being formed based off the students interests in working with specific themes. A form will be sent out for students to rank themes of interest, and will then be sorted into groups of three through their choices.

Participation (10%)

Completion of Peer Feedback and Marking (4%). The majority of the assignments will be assessed through peer marking, either through Canvas or through comPAIR. To ensure that you are equipped to provide constructive, effective, and fair feedback for your classmates, we will be spending some time in class discussing how to provide effective feedback and how to approach peer marking.

Participation in-class and through Canvas discussions (up to 3 classes may be excused throughout the semester and participation will be tracked through a check-out question or participation in an activity after each discussion) - (6%)

Late assignments will lose 10% for each day that they are handed in late, until incomplete. If there are extenuating circumstances which prevent completion, please discuss this with us.

LEARNING ANALYTICS

To foster a learning environment where all individuals feel respected and have a sense of belonging and inclusion, we will provide an anonymous feedback survey that will be kept open throughout the course for students to provide feedback. The analytics data will be used to:

- Build a supportive learning environment for students
- Review statistics on course content being accessed to support improvements in the course
- Assess your participation in the course

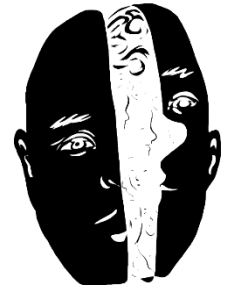
LEARNING RESOURCES

Throughout this course, students will acquire a basic understanding of neuroanatomy and develop the ability to read and analyze neuroscientific literature included in the course reading list. To support this, students will have access to foundational Canvas modules at the start of the course, which they can complete at their own pace to build a background knowledge in neuroscience.

OTHER COURSE POLICIES

Copyright. All materials of this course (course handouts, lecture slides, assessments, course readings, etc.) are the intellectual property of the Course Instructor or licensed to be used in this course by the copyright owner. Redistribution of these materials by any means without permission of the copyright holder(s) constitutes a breach of copyright and may lead to academic discipline. Classes are not permitted to be recorded in any medium.

Academic Integrity & Generative AI. The use of generative AI (e.g., ChatGPT) for the completion of reflections or written assignments is allowed, with the condition that you must include the prompt you used and the AI's output in a separate document and demonstrate how you fact-checked the information and how you used that information to build your knowledge. Using track changes and comments in Microsoft Word or suggesting mode and comments in Google Docs may be helpful here. A simple copy-paste from an AI chatbot is not acceptable and will be considered academic misconduct. When using generative AI as a tool for your learning, please be transparent in your use. If you are ever unsure if your usage of AI aligns with course or university policies, please speak with either of us.



UNIVERSITY POLICIES

UBC provides resources to support student learning and to maintain healthy lifestyles but recognizes that sometimes crises arise and so there are additional resources to access including those for survivors of sexual violence. UBC values respect for the person and ideas of all members of the academic community. Harassment and discrimination are not tolerated nor is suppression of academic freedom. UBC provides appropriate accommodation for students with disabilities and for religious observances. UBC values academic honesty and students are expected to acknowledge the ideas generated by others and to uphold the highest academic standards in all their actions.

Details of the policies and how to access support are available on [the UBC Senate website](#).